

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Implementation of Baseline Model 1(Classification Approach)

Introduction

Baseline Model 1 was developed using a traditional machine learning approach for sentiment classification. The objective was to establish a reference performance using standard text processing and classification techniques.

Data Preparation

The dataset consists of educational feedback comments labeled into three categories:

- Negative (0)
- Neutral (1)
- Positive (2)

The data was cleaned and preprocessed, followed by splitting into training and testing sets.

Feature Extraction

Text data was converted into numerical format using **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorization.

- Captures importance of words
- Includes n-grams for better context
- Limited feature size for efficiency

Model Used

A **Logistic Regression classifier** was implemented due to its effectiveness in text classification tasks.

Model Evaluation

Overall Accuracy: ~77%

Classification Report:

Class	Precision	Recall	F1-Score	Support
Negative	0.80	0.72	0.76	1209
Neutral	0.36	0.10	0.16	478
Positive	0.78	0.94	0.85	2308

Macro Avg F1: 0.59
Weighted Avg F1: 0.74

Confusion Matrix Insights

- Strong performance for **positive class**
- Moderate performance for **negative class**
- Very poor detection of **neutral class**
- Many neutral samples misclassified

Conclusion

Baseline Model 1 provides a solid starting point with good overall accuracy. However, its inability to effectively handle contextual relationships limits its performance, especially for neutral sentiment classification.